Joint shutter and linearity fit

Proposed numerical interfaces for review

Status. The paired gain/RON helper at `e0b16476` passed the actual Intel Mac C17 gate: two tests, 2.58 seconds, all three preservation checks true. Evidence: `/Users/fselman/Software/build/wenu-imaging/mac-gain-pair-AdGQvwqZ`. Issue 228 (the corrected synthetic test expectation) is closed. This note proposes the next numerical interfaces; no joint fitter is implemented.

The 2 November detector-characterization formula guide remains controlling. This review supplies the numerical contract and explicitly identifies the missing Eclipse functionality. Pixel nonlinearity thresholds, lamp-drift fitting and real-data execution are not selected here.

1. Scientific model and symbols

For a directly modeled signal in region r, exposure j and amplifier a:

$$Q_a(s) = s + c_{2,a}s^2 + c_{3,a}s^3, \tag{1}$$

$$Q_a(s_{rj}) = L_r(t_j + \delta), \qquad t_j + \delta > 0. \tag{2}$$

There is **one signed δ for the whole mosaic**, separate positive illumination rates L_r, and amplifier-specific $c_{2,a}, c_{3,a}$. There is no free additive signal offset to absorb bias error. The shutter offset is determined jointly from the accepted measurements of **all detectors**, with every detector using this same δ. Each ROI retains its own illumination rate and each amplifier its response coefficients. Excluded or saturated samples contribute no information; usable samples contribute through the full covariance-weighted objective in Section 2. We do not fit separate detector delays and then average them.

Symbol	Meaning	Unit
s_{rj}	Measured bias-subtracted signal; see aggregation rule below	ADU
$Q_a(s)$	Linearized signal for amplifier a	ADU
$c_{2,a}, c_{3,a}$	Quadratic and cubic inverse-response coefficients	ADU^{-1}, ADU^{-2}
L_r	Linearized illumination rate in region r	ADU/s
t_j	Input duration in the explicitly established time convention	s
δ	Common duration offset; positive means longer exposure	s
$\widehat{s}_i$	Predicted measured signal for observation i	ADU
$e_i = s_i - \widehat{s}_i$	Signal-space residual	ADU
C	Covariance of accepted signal measurements	ADU^2
θ	Ordered vector of fitted coefficients, rates and common offset	mixed

Pixel/ROI distinction. A nonlinear transform of a mean does not equal the mean of transformed pixels. The evaluator must predict individual pixel signals before aggregating over an ROI's accepted population. An ROI-mean approximation is permitted only with an explicit validation bound over its signal distribution and fitted interval. A table of ROI means alone does not establish that bound; an unvalidated approximation must be refused.

Known BPM pixels are excluded from reference populations according to configured bits. Saturation/nonfinite selection is per sample and independent of permanent defect classification. A saturated central ROI does not exclude unsaturated corner ROIs in that exposure. Original exposure order and pair identities remain.

2. Objective, validity and derivatives

The prediction is obtained on the validated monotonic branch:

$$\widehat{s}_{rj} = Q_a^{-1}[L_r(t_j + \delta)]\,, \tag{3}$$

$$Q_a'(s) = 1 + 2c_{2,a}s + 3c_{3,a}s^2 > 0, \tag{4}$$

$$\chi^2(\theta) = e(\theta)^{\mathsf{T}} C^{-1} e(\theta). \tag{5}$$

C is explicitly supplied and held fixed for one optimization. Diagonal weights are permitted only when justified. Shared master-bias errors, shared reference measurements and overlapping ROIs must be represented or removed by using an independent nonduplicated population. Do not silently replace C by its diagonal.

For independent observations with known standard uncertainties σ_i:

$$\chi^2 = \sum_i \left(\frac{e_i}{\sigma_i}\right)^2.$$

A parameter-dependent noise model needs a separately reviewed likelihood and weight-update policy; it is not silently substituted for this fixed-C objective. Unknown covariance cannot be relabeled as a measured uncertainty.

For a directly modeled signal, let $u = \widehat{s}_{rj}$ and $D = Q_a'(u)$. Implicit differentiation gives:

$$\frac{\partial u}{\partial \delta} = \frac{L_r}{D}, \qquad \frac{\partial u}{\partial L_r} = \frac{t_j + \delta}{D}, \qquad \frac{\partial u}{\partial c_{2,a}} = -\frac{u^2}{D}, \qquad \frac{\partial u}{\partial c_{3,a}} = -\frac{u^3}{D}.$$

The residual Jacobian is $J_{ik} = \partial e_i / \partial \theta_k$, the negative of the applicable prediction derivative. For pixel-before-ROI prediction, aggregate the prediction derivatives with exactly the same weights as the predicted signals. These equations do not authorize treating broad ROI means as single pixels.

Candidate evaluation must reject nonpositive effective durations, nonpositive illumination rates, nonmonotonic response on the full declared signal interval, and targets outside that branch. No extrapolation or silent coefficient clipping. The minimum of Q_a' on an interval can be checked at its endpoints and its interior quadratic vertex, when present. The numerical inversion must be bracketed within that same interval, with a reported convergence tolerance.

Identifiability. Constrained bias subtraction is required: an unknown bias pedestal can mimic shutter delay. Arbitrary independent lamp factors would absorb exposure-time response and are not fitted. The ten-level OCAM sequence is assessed by its usable coverage/rank, not rejected solely for having fewer than the representative annual 22 pairs.

3. Numerical scaling and uncertainty

Use positive caller-declared parameter scales d_k and dimensionless coordinates $\theta_k = d_k x_k$. Scales are numerical units, not scientific priors. They must be recorded so that rank/conditioning decisions are reproducible.

If $C = HH^{\mathsf{T}}$, with H a nonsingular lower triangular covariance factor, define whitened residuals and Jacobian:

$$\widetilde{e} = H^{-1}e, \qquad \widetilde{J} = H^{-1}J, \qquad A = \widetilde{J}\,\mathrm{diag}(d_k).$$

The input contract supplies H or an explicitly diagonal equivalent. The recipe must establish its provenance; the numerical service checks dimensions, finite entries and positive diagonal. No diagonal jitter is added to hide exact duplicate observations or singular covariance.

4. Eclipse reuse and the proposed additions

The reviewed source is the current gain/RON candidate, descended from accepted main. Existing public interfaces are preserved.

Existing resource	What it supplies	What is missing
`optimization.c: minimize`	Simplex reflection, expansion and contraction scaffolding	Checked context callback, parameter bounds/scales, finite/validity states and explicit parameter/objective convergence controls
`matrix.c: matrix_leastsq`	Matrix storage and normal-equation least squares	Pivoted QR, numerical rank and scaled condition reporting
`fit_curve.c`	Polynomial and simple response fits	Shared signed mosaic timing with constrained validity and covariance
`detector.c`	Legacy time-as-signal-polynomial fit; cubic correction primitive	The accepted joint signal-space objective and complete result contract

Addition A: checked optimizer in the existing optimization module. Proposed entry point:

```
int minimize_checked(
    const optimization_problem *problem,
    const optimization_options *options,
    optimization_result **out);
void optimization_result_del(optimization_result *result);
```

The borrowed problem contains parameter count, initial simplex, positive scales, finite bounds, a context pointer and a cost callback. The callback distinguishes a valid finite cost, an inadmissible model point and a hard evaluation failure. Options explicitly set evaluation limits, parameter-spread tolerance and objective-spread tolerance. An initial feasible simplex is required.

Extend the existing simplex scaffolding; do not introduce another optimizer inside a recipe. Invalid trial points are rejected rather than clipped. Convergence requires feasible vertices and both declared spread checks. Return an explicit nonconvergence state with diagnostics when limits are reached. A converged simplex is a candidate minimum, not evidence of a unique or scientifically identified solution. Repeated starts/profile checks belong to the detector wrapper. Keep the old `minimize` behavior/API intact.

Addition B: checked least-squares/QR diagnostic in the matrix module.

```
int matrix_leastsq_qr_checked(
    const matrix *design, const matrix *rhs,
    const double *column_scales, double rank_tolerance,
    matrix_qr_result **out);
void matrix_qr_result_del(matrix_qr_result *result);
```

Matrices follow existing row-major storage and dimensions. The design is M observations by P parameters, with $M \geq P$; the first increment accepts one M-element right-hand side. Positive column scales and a relative rank tolerance are explicit. Use column-pivoted Householder QR on the scaled design,

$$A\Pi = UR,$$

where Π is the column permutation, U has orthonormal columns and R is upper triangular. Report pivot order, rank, residual norm and a clearly labeled scaled condition estimate; do not call a diagonal ratio an exact condition number. No normal-equation inversion, hidden ridge term or substituted zero solution.

5. Detector wrapper and fit acceptance

After the two numerical interfaces are reviewed, the detector-level entry point is:

```
int detector_joint_response_fit_checked(
    const detector_joint_problem *problem,
    const detector_joint_options *options,
    detector_joint_result **out);
void detector_joint_result_del(detector_joint_result *result);
```

The borrowed problem binds amplifier/ROI/exposure identities, signal units, accepted samples, prediction/aggregation evaluator, fixed covariance factor and parameter layout. Its evaluator must implement Section 1's pixel-before-ROI contract or supply the validated restricted approximation. Compact measurement blocks and bounded image access are permitted; do not retain all detector cubes simultaneously. The wrapper supplies the shared signed offset, cubic model, analytic derivatives and validity checks to the reusable numerical routines.

Options explicitly set parameter scales/bounds, accepted signal intervals, initial simplexes, convergence/rank/conditioning limits and profile settings. No numerical scientific defaults or automatic polynomial-degree selection are introduced by this proposal.

The result owns fitted parameters with units/identities, full parameter covariance when available, residuals, accepted/excluded measurement references, valid signal intervals, rank/condition diagnostics, evaluation counts, bound/convergence states and shutter-profile diagnostics. Inputs remain immutable; general allocation/invalid-input errors leave an occupied or caller-owned output unchanged. Dedicated destructors release every result allocation. Numerical/scientific unavailability is explicit and is not published as a valid zero shutter delay.

For known absolute covariance and a locally identifiable interior solution:

$$\mathrm{Cov}(\theta) \simeq \left(J^{\mathsf{T}} C^{-1} J\right)^{-1}.$$

Compute this local linear approximation through QR/triangular solves, not by forming the normal equations. Transform back from the dimensionless parameter scales and label it as a local approximation. Do not multiply by reduced χ^2 when C already expresses absolute measurement variance. Relative-weight fitting, if requested later, needs its own scale/dof convention.

An optimizer stop is insufficient. Require adequate signal coverage, full rank, acceptable scaled conditioning, valid monotonic branches and consistent feasible solutions from declared starts. Profile the shutter term by holding δ fixed and refitting the remaining parameters:

$$\chi^2_{\mathrm{profile}}(\delta) = \min_{\theta\,:\,\theta_\delta = \delta} \chi^2(\theta).$$

A flat/unbounded profile, boundary-truncated interval or incompatible minima means the shutter is undetermined. Store the profile grid and the declared likelihood convention; do not manufacture a symmetric error bar.

BPM classification remains separate. A correctable amplifier curve does not flag every pixel. A later per-pixel residual test may set reason bit 16 only with adequate coverage and reviewed thresholds. Insufficient information is coverage-unavailable, not a permanent defect. Known BPM pixels can still undergo individual defect tests without joining the reference population.

6. Proposed implementation and verification order

1. Checked QR and ownership/rank controls in Eclipse; independent small matrix solutions, scaled/permuted columns, near-degeneracy, overflow and allocation failures.

2. Checked extension of the existing optimizer; independently known minima, bounds/inadmissible callbacks, parameter scaling, convergence limits and ownership/failure checks. Preserve legacy callers with focused comparisons.

3. Detector wrapper and analytic derivative comparisons; recover negative, zero and positive common shutter offsets and injected amplifier curvature. Exercise degeneracy, shared-bias covariance, overlapping ROI treatment, saturation, disconnected coverage and lamp-drift violations.

Announce the purpose, synthetic inputs, numerical calls and pass criteria before execution. Mac C17 first; representative Chapman validation follows for changed code before full actual-data testing there. Reuse qualified gain/RON evidence; do not rerun unchanged tests. No installation, real-data processing, merge or branch cleanup follows from publishing this proposal.

Review requested. Approve or amend the two reusable numerical interfaces and their contracts before implementation. The detector wrapper remains conditional on an explicit pixel/ROI evaluator and covariance preparation contract; this note does not claim those inputs already exist in the current BPM service. The earlier accepted scientific model is unchanged.

Teaching appendix: simplex and QR

Two different jobs. Simplex searches for a minimum of the nonlinear objective. QR solves linear least-squares problems and examines the local independence of parameter effects. QR is not a second nonlinear optimizer.

A. Simplex: searching without derivatives

For P fitted parameters, a simplex consists of $P + 1$ trial parameter vectors: a triangle in two dimensions, a tetrahedron in three. Here "simplex" means the Nelder–Mead downhill search, not the simplex algorithm of linear programming. At each vertex we evaluate the same covariance-weighted residual cost χ^2. A lower cost means a better match to the measurements.

The method orders the vertices by cost and forms the centroid of the best P. It reflects the worst vertex through that centroid. A promising reflection can be extended by expansion; an unsuccessful one leads to contraction. If that fails, the simplex shrinks toward its best vertex. These operations adapt the location and size of the search without requiring analytic derivatives.

Parameter scales make timing, illumination and curvature steps comparable in dimensionless coordinates. Bounds and physical checks reject inadmissible trial points. Small vertex separation and small cost spread are stopping conditions, not proof of a unique minimum. Repeated starts and the shutter profile are therefore required. A nearly flat direction can leave the delay undetermined even when the search stops successfully.

B. QR: separating independent parameter effects

Near a fitted point, the residual changes approximately linearly with a small parameter change. After covariance whitening and parameter scaling, write

$$\widetilde{e}(x + \Delta x) \simeq \widetilde{e}(x) + A\Delta x, \qquad \min_{\Delta x} \|A\Delta x - b\|^2, \quad b = -\widetilde{e}(x).$$

Here x denotes dimensionless parameters, Δx their small change, A the scaled, whitened residual Jacobian and b the required residual change. The double bars denote Euclidean length. Householder reflections rotate this linear problem without changing lengths. With column pivoting,

$$A\Pi = UR, \qquad Rz = U^{\mathsf{T}}b, \qquad \Delta x = \Pi z.$$

Π reorders parameter columns; U has orthonormal columns; R is upper triangular; z is the change in the reordered coordinates. For full column rank, back substitution solves the triangular system. Pivoting helps expose nearly dependent parameter effects; the declared tolerance determines numerical rank. QR avoids forming $A^{\mathsf{T}}A$, whose condition number is the square of that of A for full column rank in the Euclidean norm.

Here QR checks whether shutter delay, illumination and curvature can be distinguished locally and supports covariance calculations. It cannot create missing signal coverage or resolve a physical degeneracy. A rank-deficient result must be reported as unavailable, not converted into a precise delay.